

# Sreenidhi Vedartham

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**H-1B transfer only. Approved I-140, no lottery required.**

Software/Electrical Engineer (8+ years) spanning embedded IoT hardware, cross-team technical strategy, and applied AI prototyping. I build fast, find where systems break, and turn what I learn into workflows teams actually adopt.

## EXPERIENCE

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### Software/Electrical Engineer | Ivani LLC, Greater St. Louis Area

Jan 2018 – Present

#### Engineering | Individual Contribution (IC)

- Debugged ESP32-based firmware across error codes, serial port failures, and permit-join/network-formation bugs; traced and fixed a permit-join ordering bug in network formation, working with Nordic hardware and the OpenThread stack.
- Configured and tested wireless IoT SoCs (ESP32, SiLabs, Nordic families) for Sensify functional hardware validation, including flashing and provisioning hardware across a 1,000+ node deployment.
- Contributed to Sensify's core detection algorithm, combining applied math and ML in Python (pandas, NumPy, SciPy, Matplotlib) to turn raw sensor signals into actionable occupancy output.
- Built and deployed a prompt-engineering benchmarking program to measure cost, latency, and memory tradeoffs across structured prompting approaches — cut rework ~50% and eliminated ~8 min of wasted time per query, with org-wide adoption within two weeks.
- Designed and ran functional and field testing across Sensify hardware and software, feeding findings back into requirements; diagnosed a recurring field escalation for a partner deployment to a code-level ordering bug through on-site motion-trigger testing, and resolved it.
- Contributed to design and maintenance of a ground-truth data collection pipeline (systemd service management, Python data stack: pandas, scipy, InfluxDB, Kafka).
- Tested algorithm portability across multiple chip platforms to confirm consistent performance regardless of hardware; ran multi-OS validation testing using virtual machines.
- Wrote unit and firmware tests for embedded C implementations, including validation tests comparing Python prototypes against their embedded C ports on target hardware.
- Stress-tested AI-generated guidance for embedded SDK implementations (Zigbee broadcast/RSSI APIs) against verified documentation using local LLM infrastructure, before team adoption.
- Owned technical documentation QA as a fresh user — walked internal and external docs end-to-end to catch gaps before customers or new hires hit them, alongside MRQ test coverage.
- Created algorithm and architecture diagrams in draw.io to communicate design ideas and system structure.
- Conducted technology trade studies and competitive landscape evaluation to size up new tech options before the team committed direction; hands-on testing of new microcontrollers, ESP chips, and Silicon Labs hardware as they arrived, including at-home Raspberry Pi setups to gather real-world usage data beyond the lab.
- Designed a merge request checklist standardizing the engineering review process, pitched to and adopted by the CTO, used company-wide for five years and still evolving as needs change.
- Ran a sustained cross-team engineering coaching program over 4+ years, evolving monthly focus themes and 1:1 sessions with 5+ engineers.
- Ran weekly rubber-ducking and brainstorming sessions with engineers over 6+ years; asking naive, outside-perspective questions helped surface and resolve bugs engineers had gotten stuck on.

#### Zero-Gravity Thinker | Technical Strategy & Collaboration

- Embed with teams as an outside perspective, challenging assumptions and surfacing overlooked questions before they become downstream problems.
- Sketch requirements with engineers at a helicopter-view level before they dive into detail; built the communication scaffolding — plans-of-action on Slack, worklogs, and block diagrams — to keep work followable across the team.
- Bridge between stakeholders and engineering: identifying hesitant questions before they surface as friction, raising them in standups early, and translating honest engineer feedback upward in a form the company can act on.
- Acted as a force multiplier for engineers stuck in narrow-focus debugging loops — helping them step back to high-level thinking turned bug-fighting that could stretch into days into hours or minutes.
- Authored the team's standard troubleshooting checklist for a recurring config bug class.

#### Social Media Marketing Co-Lead

- Co-led Ivani's B2B marketing operation as a hands-on writer, video editor, and visual-content creator — shaping team routines, competitor research across six companies, content strategy, and creative review across the full production pipeline (posts, carousels, demo videos, website articles, product comparisons, one-pagers, webinar recaps, executive content, and company announcements) from concept through publication and performance analysis, in collaboration with the marketing lead.
- Translated complex RF sensing, Thread, and IEEE 802.15.4 concepts into accurate, accessible B2B messaging for technical, executive, partner, and non-technical audiences.
- Co-created, populated, and maintained a marketing playbook and repeatable document-tracking workflow; reviewed LinkedIn posts for brand consistency, technical accuracy, product claims, terminology, visuals, and IP sensitivity, routing approvals to the marketing lead, engineering managers, or C-suite based on risk.
- Owned sustained GA4 analytics and OKR reporting beginning in 2024, producing monthly stakeholder audits and quarterly and annual reports to guide messaging, content priorities, and company alignment.
- Spec'ed, edited, presented, and iterated a six-variant Sensify scrollytelling experience based on C-suite feedback, including a hands-on fix to a hotspot-positioning regression in the scene code, reframing the demo around partner comprehension and real-world impact.
- Drove the design spec for an interactive cutaway-dollhouse demo of Sensify's WITH/WITHOUT behavior.
- Designed and directed a 7-piece Sensify pain-point one-pager library built for OEM product-manager intake.
- Authored the Sensify OEM Integration Whitepaper draft: architecture diagrams across cloud/edge/hybrid/on-prem topologies, built for the OEM engineering team.
- Drove the editorial QA gate on the Ivani Marketing Audit, flagging unverifiable competitor claims before exec review, and supplied the three research compilations that became its source of truth.
- Built three scenario-based CES 2023 attendee personas (media, sales, and engineering visitor archetypes) plus a demo-preparation playbook covering audience triage, conversation pacing, technical depth, and take-away materials for booth conversations.
- Supported partner and industry-organization communications by drafting announcements, adapting messaging to Ivani's voice, checking technical accuracy, selecting calls to action, producing supporting visuals, and incorporating feedback under the marketing lead.

### **Co-Founder, CEO | Embark Women, Tyler, TX**

*Jan 2017 – Apr 2022*

- Co-created Embark Women to expand professional opportunity, mentorship, and community for women; grew the community to 800+ members on Facebook, recruited and supported 150+ members with personalized career and business guidance, and delivered 20+ programs serving more than 500 participants (counted conservatively).
- Owned and co-led full-lifecycle event and community marketing: selected topics, recruited speakers and partners, created promotional campaigns and assets, managed registration and attendee communications, coordinated venues and volunteers, developed sponsorship materials, hosted programs, published recaps, and tracked participation.
- Built recurring, low-barrier community programming, including partnering with the "Coffee Outside" series, to encourage informal relationship-building and sustained local participation in the community.
- Interviewed community members, analyzed recurring questions and feedback, and adapted programming, event messaging, and community resources to emerging needs.
- Established a no-selling/no-self-promotion policy to keep the community relationship-focused; communicated leadership and organizational-status changes and implemented consent forms for photographs, interviews, and member stories.
- Designed inclusive participation practices for newcomers and people with different financial, scheduling, transportation, childcare, disability, cultural, and comfort needs while prioritizing women and underrepresented voices in events and stories.
- Managed digital community and event infrastructure using WordPress/Wix, Meetup, Eventbrite, Google Forms, Mailchimp, Canva, Hootsuite, Buffer, and Sprout Social; also supported photography, audio, livestream, and podcast production through a local community lab.

### **Electrical Engineer | Trane, Tyler, TX**

*Jun 2017 – Jan 2018*

- Wrote a multi-threaded Ruby tool that flashed firmware on 400+ microcontroller boards remotely and simultaneously, cutting deployment time and customer downtime.
- Debugged and fixed software issues in VSPD drives (C/C++). Wrote failure scenarios and tested updates against them before they hit customer environments, because finding it in the lab beats finding it in the field.
- Designed and built embedded electrical systems supporting multiple microcontroller boards. Troubleshot with Eclipse Cpp Neon, ST-Link Programmer, and Linux IDE. Hands-on soldering and onsite debugging alongside field teams.

## PROJECTS

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**TeamTrail** — Slack Agent Builder Challenge: Native Slack app using the Assistant lifecycle, Slack Real-Time Search API, and a Notion MCP integration for cited cross-tool answers. Scoped, built, and submitted solo.

**Dead Code Finder** — GitLab Transcend Hackathon: Custom GitLab Duo agent flow querying the Orbit knowledge graph for unused code, results bucketed by confidence. Built and submitted solo.

**GreenGrid AI** — AWS Breaking Barriers Virtual Challenge: End-to-end IoT energy pipeline (simulated smart meters, MQTT, AWS IoT Core, S3, Lambda, Bedrock, Amplify frontend) delivering personalized multilingual energy-saving recommendations. Built and submitted solo.

**Other hackathon builds:** a multi-agent mental-health intake system (Google ADK, Vertex AI, Firestore), a serverless therapist-notes summarization workflow (AWS Lambda, Bedrock/Claude, DynamoDB), and a production-style memory layer for LLMs (FastAPI, PostgreSQL/pgvector, Redis).

## EDUCATION

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M.S., Electrical Engineering, University of Texas at Tyler (2015–2016)

B.E., Electronics & Communications Engineering, Alpha College of Engineering (2010–2013)

## TECHNICAL ACTIVITIES

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- 6 hackathons joined, 6 Devpost projects shipped
- 2 GitHub issues filed against Google ADK (1 co-authorship invitation)

## CERTIFICATIONS & LANGUAGES

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Google AI Essentials · Google Prompting Essentials · Introduction to Generative AI Learning Path

English (Full Professional) · Tamil (Native or Bilingual)